Colloidal Particles	Shapes
As_2S_3	Spherical
Fe $(OH)_3$ sol (blue gold sol)	Disc or plate like
W_3O_5 sol (tungstic acid sol)	Rod like

10) Optical property

Colloids have optical property. When a homogeneous solution is seen in the direction of light, it appears clear but it appears dark, in a perpendicular direction.

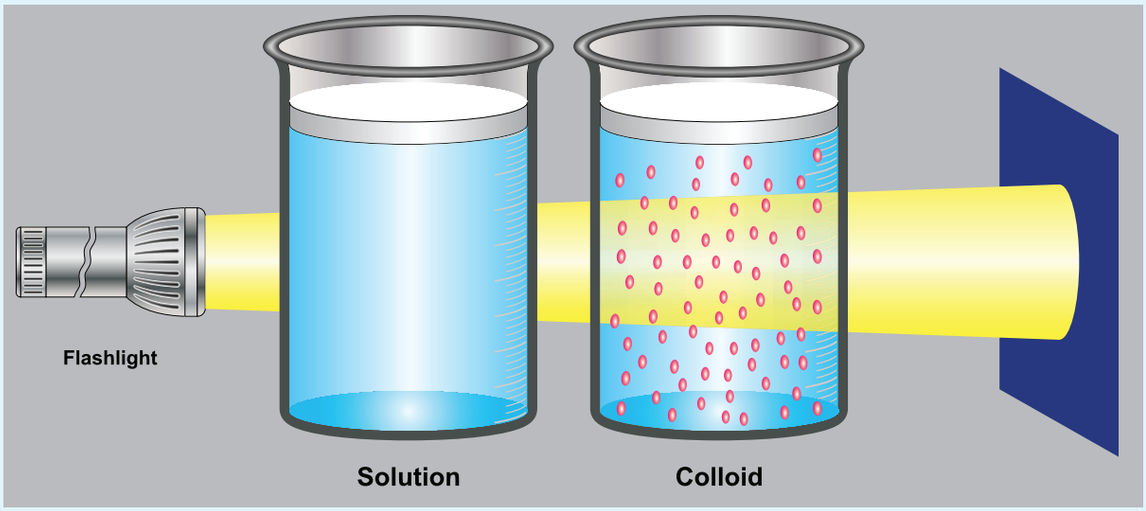


Figure 10.12 Tyndall effect

But when light passes through colloidal solution, it is scattered in all directions. This effect was first observed by Faraday, but investigations are made by Tyndall in detail, hence called as Tyndall effect.

The colloidal particles absorb a portion of light and the remaining portion is scattered from the surface of the colloid. Hence the path of light is made clear.

11) Kinetic property

Robert Brown observed that when the pollen grains suspended in water were viewed through ultra microscope, they showed a random, zigzag ceaseless motion.

This is called Brownian movement of colloidal particles.

This can be explained as follows

The colloidal sol particles are continuously bombarded with the molecules of the dispersion medium and hence they follow a zigzag, random, continuous movement.

XII U10-Surface Chemistry.indd 93 11-01-2021 15:33:44